

Hackathon @ Chips and AI

Group Name: Core 4

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In-Memory Acceleration of Monte Carlo and ML-Based Financial Workloads

Overview - Problem and Motivation

The problem we want to solve is to make intensive financial models more efficient in data movement, latency, energy consumption, and possibly throughput while maintaining accuracy. In this research, we will be using in-memory computing (IMC) to simulate financial models. We would use option pricing as the example workload. The idea is to compare a traditional Monte Carlo pricing method with a small neural-network model, and then look at how the neural-network workload could benefit from an in-memory computing architecture.

Technical Approach / Proof of Concept

1. Monte Carlo Option Pricing

First, we would implement Monte Carlo option pricing. Instead of predicting one future stock price, Monte Carlo randomly simulates thousands of possible future stock-price scenarios and uses their payoffs to estimate the option's value. This gives us a realistic but computationally intensive baseline.

2. Neural-Network Surrogate

Next, we would use the Monte Carlo results as training data for a small neural-network surrogate. The inputs could include stock price, strike price, volatility, time to expiration, and interest rate, while the output is the option price. Once trained, the model could approximate a price without rerunning thousands of Monte Carlo simulations each time. The tradeoff is that the neural network is an approximation, so there may be some pricing error.

3. In-Memory Computing

Finally, we would model how the neural network's computations could be mapped onto an in-memory or near-memory computing architecture. In a conventional system, data is repeatedly moved between memory and the processor. IMC allows some computation to happen in or close to memory, which could reduce data movement and potentially lower latency and energy use.

Theme: 4. In-Memory Computing and Circuits

Topic 4 focuses on reducing the cost of data movement through in-memory or near-memory computing and includes simulation and mapping of neural networks onto non-traditional hardware. Our project would be a software-based proof of concept rather than building a physical chip.

Evaluations

- Monte Carlo option pricing as the baseline
- Neural-network surrogate on conventional computing
- Neural-network workload mapped to a simulated IMC architecture

Measurement of Success

We could compare pricing accuracy, estimated data movement, latency, energy consumption, and possibly throughput. The main question is whether we can achieve computational efficiency without sacrificing much pricing accuracy.

Proposed Workflow

Monte Carlo → Training Data → Neural-Network Surrogate → IMC Mapping → Performance Comparison